

Integrated metabolic modelling reveals cell-type specific epigenetic control points of the macrophage metabolic network.

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The reconstruction of context-specific metabolic models from easily and reliably measurable features such as transcriptomics data will be increasingly important in research and medicine. Current reconstruction methods suffer from high computational effort and arbitrary threshold setting. Moreover, understanding the underlying epigenetic regulation might allow the identification of putative intervention points within metabolic networks. Genes under high regulatory load from multiple enhancers or super-enhancers are known key genes for disease and cell identity. However, their role in regulation of metabolism and their placement within the metabolic networks has not been studied. Here we present FASTCORMICS, a fast and robust workflow for the creation of high-quality metabolic models from transcriptomics data. FASTCORMICS is devoid of arbitrary parameter settings and due to its low computational demand allows cross-validation assays. Applying FASTCORMICS, we have generated models for 63 primary human cell types from microarray data, revealing significant differences in their metabolic networks. To understand the cell type-specific regulation of the alternative metabolic pathways we built multiple models during differentiation of primary human monocytes to macrophages and performed ChIP-Seq experiments for histone H3 K27 acetylation (H3K27ac) to map the active enhancers in macrophages. Focusing on the metabolic genes under high regulatory load from multiple enhancers or super-enhancers, we found these genes to show the most cell type-restricted and abundant expression profiles within their respective pathways. Importantly, the high regulatory load genes are associated to reactions enriched for transport reactions and other pathway entry points, suggesting that they are critical regulatory control points for cell type-specific metabolism. By integrating metabolic modelling and epigenomic analysis we have identified high regulatory load as a common feature of metabolic genes at pathway entry points such as transporters within the macrophage metabolic network. Analysis of these control points through further integration of metabolic and gene regulatory networks in various contexts could be beneficial in multiple fields from identification of disease intervention strategies to cellular reprogramming.

Signal Integration by Shadow Enhancers and Enhancer Duplications Varies across the Drosophila Embryo.

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Transcription of developmental genes is controlled by multiple enhancers. Frequently, more than one enhancer can activate transcription from the same promoter in the same cells. How is regulatory information from multiple enhancers combined to determine the overall expression output? We measure nascent transcription driven by a pair of shadow enhancers, each enhancer of the pair separately, and each duplicated, using live imaging in Drosophila embryos. This set of constructs allows us to quantify the input-output function describing signal integration by two enhancers. We show that signal integration performed by these shadow enhancers and duplications varies across the expression pattern, implying that how their activities are combined depends on the transcriptional regulators bound to the enhancers in different parts of the embryo. Characterizing signal integration by multiple enhancers is a critical step in developing conceptual and computational models of gene expression at the locus level, where multiple enhancers control transcription together.

Rabbit milk protein genes: from mRNA identification to chromatin structure.

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Milk protein genes are among the most intensively expressed and they are active only in epithelial mammary cells of lactating animals. They code for proteins which represent 30% of the proteins consumed by humans in developed countries. Mammary gland development occurs essentially during each pregnancy. This offers experimenters attractive models to study the expression mechanisms of genes controlled by known hormones and factors (prolactin, glucocorticoids, progesterone, insulin-like growth factor-1 and others) as well as extracellular matrix. In the mid-1970s, it became possible to identify and quantify mRNAs from higher living organisms using translation in reticulocyte lysate. A few years later, the use of radioactive cDNAs as probes made it possible for the quantification of mRNA in various physiological situations using hybridisation in the liquid phase. Gene cloning offered additional tools to measure milk protein mRNAs and also to identify transcription factors. Gene transfer in cultured mammary cells and in animals contributed greatly to these studies. It is now well established that most if not all genes of higher eukaryotes are under the control of multiple distal regulatory elements and that local modifications of the chromatin structure play an essential role in the mechanisms of differentiation from embryos to adults. The technique, known as ChIP (chromatin immunoprecipitation), is being implemented to identify the factors that modify chromatin structure at the milk protein gene level during embryo development, mammary development and lactogenesis, including the action of hormones and extracellular matrix. Transgenesis is not just a tool to study gene regulation and function, it is also currently used for various biotechnological applications including the preparation of pharmaceutical proteins in milk. This implies the design of efficient vectors capable of directing the secretion of recombinant proteins in milk at a high concentration. Milk protein gene promoters and long genomic-DNA fragments containing essentially all the regulatory elements of milk protein genes are used to optimise recombinant protein production in milk.

Functional characterization of QT interval associated SCN5A enhancer variants identify combined additive effects.

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Several empirical and theoretical studies suggest presence of multiple enhancers per gene that collectively regulate gene expression, and that common sequence variation impacting on the activities of these enhancers is a major source of inter-individual variability in gene expression. However, for vast majority of genes, enhancers and the underlying regulatory variation remains unknown. Even for the genes with well-characterized enhancers, the nature of the combined effects from multiple enhancers and their variants, when known, on gene expression regulation remains unexplored. Here, we have evaluated the combined effects from five SCN5A enhancers and their regulatory variants that are known to collectively correlate with SCN5A cardiac expression and underlie QT interval association in the general population. Using small deletions centered at the regulatory variants in episomal reporter assays in a mouse cardiomyocyte cell line we demonstrate that the variants and their flanking sequences play critical role in individual enhancer activities, likely being a transcription factor (TF) binding site. By performing oligonucleotide-based pulldown assays on predicted TFs we identify the TFs likely driving allele-specific enhancer activities. Using all 32 possible allelic synthetic constructs in reporter assays, representing the five biallelic enhancers in tandem in their genomic order, we demonstrate combined additive effects on overall enhancer activities. Using transient enhancer assays in developing zebrafish embryos we demonstrate the four out of the five enhancer elements act as enhancers in vivo. Together,

these studies extend the previous findings to uncover the TFs driving the enhancer activities of QT interval associated SCN5A regulatory variants, reveal the additive effects from allelic combinations of these regulatory variants, and prove their potential to act as enhancers in vivo .

Conserved and non-conserved enhancers direct tissue specific transcription in ancient germ layer specific developmental control genes.

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Identifying DNA sequences (enhancers) that direct the precise spatial and temporal expression of developmental control genes remains a significant challenge in the annotation of vertebrate genomes. Locating these sequences, which in many cases lie at a great distance from the transcription start site, has been a major obstacle in deciphering gene regulation. Coupling of comparative genomics with functional validation to locate such regulatory elements has been a successful method in locating many such regulatory elements. But most of these studies looked either at a single gene only or the whole genome without focusing on any particular process. The pressing need is to integrate the tools of comparative genomics with knowledge of developmental biology to validate enhancers for developmental transcription factors in greater detail. Our results show that near four different genes (*nkx3.2*, *pax9*, *otx1b* and *foxa2*) in zebrafish, only 20-30% of highly conserved DNA sequences can act as developmental enhancers irrespective of the tissue the gene expresses in. We find that some genes also have multiple conserved enhancers expressing in the same tissue at the same or different time points in development. We also located non-conserved enhancers for two of the genes (*pax9* and *otx1b*). Our modified Bacterial artificial chromosome (BACs) studies for these 4 genes revealed that many of these enhancers work in a synergistic fashion, which cannot be captured by individual DNA constructs and are not conserved at the sequence level. Our detailed biochemical and transgenic analysis revealed Foxa1 binds to the *otx1b* non-conserved enhancer to direct its activity in forebrain and otic vesicle of zebrafish at 24 hpf. Our results clearly indicate that high level of functional conservation of genes is not necessarily associated with sequence conservation of its regulatory elements. Moreover certain non conserved DNA elements might have role in gene regulation. The need is to bring together multiple approaches to bear upon individual genes to decipher all its regulatory elements.

Predicting spatial and temporal gene expression using an integrative model of transcription factor occupancy and chromatin state.

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Precise patterns of spatial and temporal gene expression are central to metazoan complexity and act as a driving force for embryonic development. While there has been substantial progress in dissecting and predicting cis-regulatory activity, our understanding of how information from multiple enhancer elements converge to regulate a gene's expression remains elusive. This is in large part due to the number of different biological processes involved in mediating regulation as well as limited availability of experimental measurements for many of them. Here, we used a Bayesian approach to model diverse experimental regulatory data, leading to accurate predictions of both spatial and temporal aspects of gene expression. We integrated whole-embryo information on transcription factor recruitment to multiple cis-regulatory modules, insulator binding and histone modification status in the vicinity of individual gene

loci, at a genome-wide scale during *Drosophila* development. The model uses Bayesian networks to represent the relation between transcription factor occupancy and enhancer activity in specific tissues and stages. All parameters are optimized in an Expectation Maximization procedure providing a model capable of predicting tissue- and stage-specific activity of new, previously unassayed genes. Performing the optimization with subsets of input data demonstrated that neither enhancer occupancy nor chromatin state alone can explain all gene expression patterns, but taken together allow for accurate predictions of spatio-temporal activity. Model predictions were validated using the expression patterns of more than 600 genes recently made available by the BDGP consortium, demonstrating an average 15-fold enrichment of genes expressed in the predicted tissue over a naïve model. We further validated the model by experimentally testing the expression of 20 predicted target genes of unknown expression, resulting in an accuracy of 95% for temporal predictions and 50% for spatial. While this is, to our knowledge, the first genome-wide approach to predict tissue-specific gene expression in metazoan development, our results suggest that integrative models of this type will become more prevalent in the future.

Large-scale analysis of the integration of enhancer-enhancer signals by promoters.

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Genes are often regulated by multiple enhancers. It is poorly understood how the individual enhancer activities are combined to control promoter activity. Anecdotal evidence has shown that enhancers can combine sub-additively, additively, synergistically, or redundantly. However, it is not clear which of these modes are more frequent in mammalian genomes. Here, we systematically tested how pairs of enhancers activate promoters using a three-way combinatorial reporter assay in mouse embryonic stem cells. By assaying about 69,000 enhancer-enhancer-promoter combinations we found that enhancer pairs generally combine near-additively. This behaviour was conserved across seven developmental promoters tested. Surprisingly, these promoters scale the enhancer signals in a non-linear manner that depends on promoter strength. A housekeeping promoter showed an overall different response to enhancer pairs, and a smaller dynamic range. Thus, our data indicate that enhancers mostly act additively, but promoters transform their collective effect non-linearly.

Dynamic model for the coordination of two enhancers of broad by EGFR signaling.

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Although it is widely appreciated that a typical developmental control gene is regulated by multiple enhancers, coordination of enhancer activities remains poorly understood. We propose a mechanism for such coordination in *Drosophila* oogenesis, when the expression of the transcription factor Broad (BR) evolves from a uniform to a two-domain pattern that prefigures the formation of two respiratory eggshell appendages. This change reflects sequential activities of two enhancers of the *br* gene, early and late, both of which are controlled by the epidermal growth factor receptor (EGFR) pathway. The late enhancer controls *br* in the appendage-producing cells, but the function of the early enhancer remained unclear. We found that the early enhancer is essential for the activity of the late enhancer and induction of eggshell appendages. This requirement can be explained by a mechanism whereby the BR protein produced by the early enhancer protects the late enhancer from EGFR-dependent repression. We illustrate this complex mechanism using a computational model that correctly predicts the wild-type

dynamics of BR expression and its response to genetic perturbations.

N-acetylcysteine infusion in viral myocarditis: a case report.

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N-acetylcysteine improves survival in established acute liver failure following paracetamol overdose by reducing the incidence of multiorgan failure. These benefits are thought to be related to decreased tissue hypoxia by the enhancement of both oxygen delivery and oxygen extraction. Similar findings have been recorded in critically ill patients from an alternative aetiology. The cardiovascular properties of N-acetylcysteine are to increase stroke volume index, and thus cardiac output, although there is no effect on cardiac output in normal subjects. N-acetylcysteine is known to improve myocardial contraction in a hamster model of chronic myocardial ischaemia, but such effects have not previously been described in humans. We report the beneficial circulatory effect of N-acetylcysteine in a patient with marked left ventricular dysfunction secondary to acute viral myocarditis.

Intravenous acetylcysteine in paracetamol induced fulminant hepatic failure: a prospective controlled trial.

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To see whether intravenous acetylcysteine would improve outcome in patients with fulminant hepatic failure after paracetamol overdose. A prospective randomised controlled study. The Institute of Liver Studies, King's College Hospital, London. 50 consecutive patients (21 male) aged 16-60 with fulminant hepatic failure after paracetamol overdose who had not previously received acetylcysteine. Conventional intensive liver care plus either acetylcysteine (25 patients) in the same dose regimen as used early after a paracetamol overdose, except that the infusion was continued until recovery from encephalopathy or death, or an equivalent volume of 5% dextrose (25 patients). Survival; incidence of cerebral oedema, renal failure, and hypotension requiring inotropic support; liver function as assessed by prolongation of the prothrombin time; and degree of encephalopathy. The rate of survival was significantly higher in the acetylcysteine treated group than in the controls (48% (12/25 patients) v 20% (5/25); $p = 0.037$, 95% confidence interval for difference in proportions surviving 3% to 53%). Acetylcysteine treated patients had a lower incidence of cerebral oedema (40% (10/25) v 68% (17/25); $p = 0.047$, 95% confidence interval for difference in incidence 2% to 54%), and fewer developed hypotension requiring inotropic support (48% (12/25) v 80% (20/25); $p = 0.018$, 95% confidence interval 7% to 57%). Rates of deterioration and recovery of liver function, however, were similar in the two groups. No adverse reactions to acetylcysteine were seen. Acetylcysteine is safe and effective in fulminant hepatic failure after paracetamol overdose.
